Rupali Bhatnagar

Senior Software Engineer

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Work Experience

ServiceNow	Sr. Software Engineer (Security Operations)	Mar'22 – May'25

Individual contributor on Security Operations and Vulnerability Response, building on the ServiceNow© platform. Underlined features link to ServiceNow's public documentation.

- **Exception Management — Approval Rule Engine** for vulnerability exceptions — designed a multi-level, dynamic approval rule engine on ServiceNow Flow Designer supporting deferment and exception requests across differing enterprise policies; Flow Designer approvals shipped in **v15.0.2**. **Reduced approval time by 40%.**
- **Penetration Testing Request Catalog & Workflow Automation** — built the request catalog and automated intake, triage and routing for penetration-test requests across **50+ subsystems**, **cutting manual coordination by more than 30%** — the testing itself stays expert and manual by design.
- **Background Job Processing Framework** — architected the multithreaded framework for long-running tasks that previously strained system resources, partitioning work into parallel chunks, with migration via a configuration interface. Rollup jobs were converted to multithreaded background jobs in **v18.2.5**:
  - Multithreaded execution — **60% efficacy improvement** in some use cases, **1M+ tasks/month**.
  - Job status tracking and cancellation — **100% visibility** on executing and errored tasks.
  - Configurable **retry logic and dead-letter queuing** for failed jobs.
- **Real-Time Data Aggregation Engine** — high-performance aggregation framework for near real-time multi-dimensional analytics, powering key internal dashboards over **100K+ records in an 8-hour window**.

ServiceNow	Software Engineer (Security Operations)	Apr'19 – Mar'22

- **Assignment Rule Engine for Vulnerability Remediation** — cache-optimised engine prioritising infrastructure vulnerabilities and policy violations and routing **vulnerable items (VITs)** to the correct remediation groups via the Glide platform. Used daily by ITIL users; handles **1M+ vulnerable items daily**.
- **Assignment Recommendations on Predictive Intelligence** — classification models on the **Predictive Intelligence** framework with feature engineering and a reviewer-facing recommendation interface across **vulnerable items and vulnerability groups**; shipped in **v12.1.4**. **20% increase in task allocation accuracy.**
- **CISO & Vulnerability Dashboards** — led design and development of the CISO dashboard and supporting vulnerability dashboards, defining the key metrics and structuring the data pipeline. Used on a regular basis by **50+ organisations**.
- **Vulnerability Source Integration** — engineered integrations with **Veracode, Qualys, Tenable and Microsoft TVM**, building the **ETL pipelines** that standardise **500K+ vulnerability records/day** into a common model, with error handling and retry for reliable ingestion into downstream remediation workflows.
- **POA&M Generator** — auto-generated **Plans of Action & Milestones**, the remediation-plan artifact required under **NIST SP 800-53** and **FedRAMP**, directly from vulnerability findings rather than from control failures. Built at an internal hackathon and **adopted by the Now on Now support team**, who ran it against ServiceNow's own instance.
- **Hackathon Projects**:
  - **Diagnostic Framework** — internal support tooling that streamlined issue tracking and reproduction across customer instances, **reducing debugging time by 40%**.
  - **ML over Vulnerability Response data** — Predictive Intelligence **clustering** to group vulnerable items beyond the product's filter-based group rules, and **regression** to predict time-to-remediate, where shipped remediation targets are deterministic date arithmetic.
  - **High-Volume Data Generator** — scalable generation of **100M+ records across 20–30 tables** for load testing; used by the performance team.

Darwinbox Digital Solutions	Software Developer	Aug'18 – Mar'19

- Designed and implemented a microservice architecture capturing an **audit trail** of client data modifications, written in **PHP with the Yii framework**, listening to **MongoDB oplogs** and publishing to an **AWS SQS** queue.
- Built an **Oplog Processor in Node.js** diffing strings, arrays and embedded documents into human-readable audit entries, with an internal dashboard for adding trackable fields.

NowFloats (Boost360)	Software Developer	Jun'17 – Jul'18

- Developed responsive web features in **C#, Bootstrap, JavaScript, jQuery and Knockout.js**, used daily by **50,000+ clients**.
- Integrated the **Facebook Graph API** so merchants could publish products and updates to their Facebook profile, page and shop (Ads extension).
- Built the **SPIDER** module using **machine learning** to predict sales revenue per representative from historical performance, with up to **84% accuracy**.
- Built an **offers and discounted-purchase module** over **Boost Keyboard**, integrating the internal **Kitsune**, **k-Pay** and **Assured Purchase** systems.
- Designed a **generic Booking System** for hotels, clinics and restaurants using **Node.js** and the WebActions API, and an **Order Management** dashboard (**AngularJS, C#**) for purchases, payment and delivery status and grievance redressal.

Finomena (ceased operations May'17)	**Software Developer**	**Feb'17 – May'17**

- Built internal dashboards for **Loan Management, Repayments, Collections and Transaction Tracking** in **PHP (Laravel)**, scaling to **8,000+ loans/month** with Eloquent ORM over MySQL.
- Used **Artisan** for task scheduling, cron handling and testing; Laravel modules for authorisation, validation, event handling and mailing.

Babajob.com (acquired by Quikr, Jun'17)	**Software Engineer**	**May'16 – Jan'17**

- Redesigned the **telephony system (IVR and SMS)** to scale to **10,000+ inbound calls and 20,000+ outbound SMS per day**, built in **Golang** with **RabbitMQ** as the message bus, **Redis** as in-memory cache and **MongoDB** for persistence.
- Designed and implemented RESTful outbound SMS services and a basic logging framework in Golang for debugging and tracing.
- Contributed to the **search subsystem in Java over Elasticsearch**, with dynamic query generation via Mustache templates.

Selected Projects — 2025 to present

Independent and consulting engineering through **Tathya Dev** (tathya.dev) — **10+ production web products across 43 repositories and 4,238 commits in 12 months**, enabled by AI-augmented engineering workflows.

AI & agent tooling

propagate Claude Code plugin · Node, ESM	Sole-authored: **218 commits over six weeks**, 313 source files, **142 tests** — github.com/Rupali59/propagate. Declares the couplings between files that must change together and derives drift from file *content* on demand rather than storing state, currently watching **~920 edges across 14 workspaces**. Installed at user level, so its hooks execute in **every agent session on the machine** rather than only where it is checked out. It never edits a downstream file: the tool finds and explains, a person approves, and the decision is recorded as a ledger event. A companion marketplace serves **18 skills across 3 plugins**.
Agent Guardrail & Rule System PreToolUse / SessionStart hooks	A `PreToolUse` hook pattern-matches the literal command an agent is about to execute against declared hazards and surfaces the matching one *before* it runs; written after a documented hazard sitting three files away was hit anyway and cost **11 false verification records**. A companion `SessionStart` hook maintains **18 canonical engineering rules**, and a restatement detector tracks them across **50 `CLAUDE.md` files** — it counts a file that restates a rule *even where the file also references it*, because a pointer beside a stale copy is what a half-finished migration looks like.
LLM Classification Pipeline Gemini · schema-constrained	Multi-source ingestion scoring with Gemini calls constrained by enforced response schemas, run at `temperature: 0` and batched, with determinism resting on cache-on-first-write rather than on the model, and a keyword-classifier fallback so a provider outage degrades quality instead of breaking ingestion. Model-proposed sources are **existence-checked against the live API before acceptance**, and an inconclusive check returns the candidate flagged rather than asserted.
Evaluation Harnesses selftests · fixtures	The rule detectors and both pattern-matching guardrails each ship a `selftest` that runs each trigger against a known-positive and asserts it fires, plus negative cases it must *not* match, because **a check that cannot fail reports success**; it caught a rule whose detector could never have matched its own body. A Claude Cowork job-classification skill is tested against captured fixtures.
Human-in-the-Loop Design append-only ledger	No downstream file is ever edited automatically — the tooling finds and explains, a person approves, and each decision is recorded as a ledger event. The design principle throughout is that **"found nothing" and "looked at nothing" must be different outputs** — a probe that cannot report its own failure reports success. Auditing my own monitor against that standard found it logging a hardcoded success field while **49 crashes sat in a stderr log nothing read**; the logger could not express the state it was being cited to prove.

Products & platforms

Multi-tenant CRM & Workflow Platform Shree Shyam Jyotish Kendra (Pt. Pawan Kaushik) · ssjk-crm.vercel.app	Express API plus two Next.js 14 applications (client and admin). Built a **database-driven RBAC/ACL engine** — runtime-editable roles, generic field-level access control over CRUD, per-actor tier projection, table→field cascade, and a capability-matrix UI whose review-diff modal shows the consequences of a change before an atomic batch save, with fail-closed reconciliation and admin-floor guards. WebAuthn + OTP authentication, BullMQ/Redis job queue, workflow engine, web-push. Phased task-model migration dark-shipped in five stages with a one-shot backfill of **~14,000 records**. Test infrastructure cut from **104s to 28s** across two changes — one shared `mongod` for the suite instead of 89, then per-suite index scoping, which is what actually moved the number. **228 test files.**
Content & SEO Platform Vipin Kaushik · vipinkaushik.com	Production Next.js site backed by the computation service below — the two are one system. Nine templated landing pages plus a hub on a shared shell with **FAQPage structured data** and selective `noindex`, a server-side analytics pipeline, signed-cookie location handling with a durable resolved-location cache, and a migration of public pages to **static generation**. Largest single build of the last quarter.

Demand-Intelligence Platform Next.js, TypeScript, Mongoose, Gemini	Multi-source ingestion and opportunity scoring over Trends, YouTube and Search Console, fed by the schema-constrained Gemini classification pipeline described above, with an entity ontology mapping demand against supply. Search Console ingestion authenticated via **Workload Identity Federation**, adopted to remove service-account key fragility; scheduling moved off hosted cron onto a local agent.
Ephemeris Computation Service Python 3.12, FastAPI, MongoDB	FastAPI service over Swiss Ephemeris, a native library with process-global mutable state that is **not thread-safe**. Every call is serialised through a single-worker executor that releases the event loop during computation. Adds an astronomical-timeline subsystem (boundary and instant finders, lunar-month tables, per-year caching with TTL) and typed per-calculation endpoints replacing one overloaded route. **1,034 tests.**
Motherboard Go 1.25, Next.js, Docker	Go multi-module monorepo implementing an API gateway and multi-tenant plugin platform — **25 `go.work` modules, 652 tracked Go files**. Plugin services authenticate, publish their routes to a client manager, and the gateway proxies to them behind a **fail-closed entitlement gate** and a plugin-key claim assertion.
E-commerce Storefront & Embedded Admin Gemastrology · Shopify, Liquid, Remix, Prisma	Shopify Liquid theme plus an embedded Remix admin application — metafield/metaobject schema installer, headless catalogue seeders, and certificate verification implemented natively in Liquid rather than through an app proxy. Accessibility pass to a **44×44 touch-target floor**.
Production Portfolio Sites Manav Daehi · Kirti Kumar · Youvan Prakashan	manavdaehi.com — Google Docs→works sync pipeline and multi-codec delivery (AV1/HEVC/H.264); a hero codec ladder took **10.4 MB to ~3 MB**, with delivery suppressed for non-watchers and an H.264 Level 4.0 re-encode for iOS autoplay. kirtik.co.uk — dual-track portfolio with Playwright end-to-end coverage. rtistic.vercel.app — Next.js 14 and MongoDB, jest suite and CI, public source. youvanprakashan.org — publishing platform with an Express admin classification queue and a language classifier.
Git Forensics & Recovery Incident work	A worktree removal combined with dropped stashes left **154 commits unreachable** in a live repository. All 154 were located, recovered and permanently tagged under `refs/tags/rescue/*`; **8 of 29 stashes** held content present in no branch, and the restored work was reintegrated behind its tests. Separately diagnosed a repository whose `origin` pointed at an **unrelated project with no common ancestor** — 48 commits and 341 files that a single `git push` would have sent to the wrong remote.
Text Digitisation & Translation Pipeline Python · github.com/Rupali59/Sanskrit-texts	Ingestion for a structured-JSON corpus of classical texts: a TEI/SARIT parser handling prose and division-type detection, a scraper splitting on the source's own headings, a verse-marker grammar, and per-work structural validators. One structural fix took depth errors in a single work from **3,522 to 6**. Corpus spans **90,000+ verses across 50+ texts**.

Dissertation & Coursework

Dissertation	Human Activity Recognition using an **HMM-based Intermediate Matching Kernel**, representing videos as sequences of sets of feature vectors. Re-implemented in 2026 as a containerised Python pipeline with a Flask front end — github.com/Rupali59/himk-implementation.
Methodology	Classifies a video into a predefined activity class. Feature extraction and classification in Python with scikit-learn, hmmlearn and OpenCV; originally prototyped in MATLAB with Piotr's CV Toolbox, PNetCDF and OpenMPI.
ML Techniques	Bayesian classifiers, Gaussian mixture models, perceptrons, artificial neural networks and support vector machines on linearly separable, non-linearly separable and overlapping data.
Presentations	Presented the components of an information retrieval system and Google's PageRank model at the departmental seminar on Information Retrieval.
Other Projects	Management Information System (MIS), Cybernetics Protector System, Product Management System.

Technical Skills

Languages	Go, TypeScript, JavaScript, Python, PHP, Java, C#, C++, GlidePlatform
Web & Frameworks	Next.js, React, Node.js, Express, Remix, Astro, Laravel, .NET, FastAPI, Tailwind, Bootstrap
AI & Agent Tooling	Claude Code plugins, hooks and skills; structured LLM output and schema-constrained generation; eval harnesses and fixture-based regression suites; hallucination containment and graceful degradation; human-in-the-loop design; MCP
Databases	MongoDB, PostgreSQL, MySQL, Redis, Elasticsearch
Queues & Messaging	RabbitMQ, BullMQ, AWS SQS, Redis Queue
Cloud & DevOps	AWS, Vercel, Fly.io, Docker, Shopify, Jenkins, Git, Linux
ML	scikit-learn, hmmlearn, OpenCV, MATLAB/Octave, pattern recognition and classification

Academic Profile

Year	Degree	Discipline	Institution	Score
2016	**M.Tech**	Computer Science & Engineering Pattern Recognition & Machine Learning	National Institute of Technology, Goa	8.42 CGPA

2014	**B.E.**	Computer Science & Engineering Graduated with Honours	LNCT, Bhopal (RGPV)	8.58 CGPA
2010	XII (CBSE)	Science	St. Joseph's Convent School, Bhopal	83.0%
2008	X (CBSE)	—	St. Joseph's Convent School, Bhopal	88.0%

Certifications

AI — 2026	**Anthropic:** Claude Code in Action; Introduction to Claude Cowork; Claude Platform 101; Claude Code 101; Claude 101. **Google:** Gemini Enterprise Agent Ready; Create Your First Gemini Enterprise Application; Google Cloud Innovator; Developer Program Premium Tier; AI Fundamentals (Google Career Certificates). **Verify:** credly.com/users/rupali-bhatnagar-59 · g.dev/rupali59.
Earlier	WhatsApp API Expert, Wati (2024) · Ultimate JavaScript: Fundamentals, Code With Mosh (2025) · Neural Networks & Deep Learning, Coursera (2020) · JavaScript Algorithms & Data Structures, freeCodeCamp (2020) · Git Essential Training, LinkedIn Learning (2020)

Achievements & Activities

Professional	• Winner, NowFloats Hackathon for product innovation in sales — built the ML-powered SPIDER revenue-prediction module. 2018 • **Geektrust GOLD** — score **95.8** on the Metro Card backend challenge, with **Readability** and **Maintainability** assessed at **ADVANCED** tier by an independent reviewer.
Academic	• **Chancellor's Scholarship (Topper)**, Rajiv Gandhi Proudyogiki Vishwavidyalaya, Bhopal — awarded to the top 10 students statewide by theory marks. 2012, 2013 • **Silver Medal**, Persistent Systems Limited Programming Contest, inter-college. 2013
Other	• Organised the intra-college programming competition, LNCT Bhopal. 2012 • Conducted cultural events, NIT Goa. 2014–2016

Positions of Responsibility

NIT Goa	Teaching Assistant, Computer Science & Engineering Department	2014–2016
NIT Goa	Vice President, Computer Science Department	2014–2016
LNCT, Bhopal	Student Coordinator, Training & Placement Department	2013–2014